

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals	
NADH Dehydrogenase Complex Assembly (GO: -0.31854752	48	2.365e-14	6.367e-11	TMEM126A:9	NDUFS3:34	
Mitochondrial Respiratory Chain Complex	-0.31854752	48	2.365e-14	6.367e-11	TMEM126A:9	NDUFS3:34
Mitochondrial Respiratory Chain Complex	-0.22831413	76	6.325e-12	1.135e-08	TMEM126A:9	NDUFS3:34
Monoatomic Cation Transmembrane Transport	0.12226080	265	9.404e-12	1.266e-08	KCNQ3:8	KCNV2:43
Mitochondrial Gene Expression (GO:014005	-0.18121894	99	4.994e-10	5.379e-07	POLRMT:30	TFB2M:45
tRNA Methylation (GO:0030488)	-0.29393566	36	1.063e-09	9.542e-07	THADA:6	TRMT10C:27
Inorganic Cation Transmembrane Transport	0.10722450	268	1.854e-09	1.426e-06	KCNQ3:8	KCNV2:43
Potassium Ion Transmembrane Transport (G	0.15197531	129	2.748e-09	1.850e-06	KCNQ3:8	KCNV2:43
Oxidative Phosphorylation (GO:0006119)	-0.23087484	54	4.539e-09	2.716e-06	NDUFS3:34	NDUFS5:92
Chemical Synaptic Transmission (GO:00072	0.11023545	234	7.371e-09	3.969e-06	KCNQ3:8	RIMBP2:25
Proton Motive Force-Driven Mitochondrial	-0.23854919	47	1.572e-08	7.694e-06	NDUFS3:34	NDUFS5:92
Mitochondrial Electron Transport, NADH T	-0.28594444	31	3.639e-08	1.447e-05	NDUFS3:34	NDUFS5:92
Modulation Of Chemical Synaptic Transmis	0.15719355	103	3.761e-08	1.447e-05	TRIO:114	CACNG5:152
Proton Motive Force-Driven ATP Synthesis	-0.22396591	51	3.235e-08	1.447e-05	NDUFS3:34	NDUFS5:92
Aerobic Respiration (GO:0009060)	-0.21858963	52	5.096e-08	1.759e-05	NDUFS3:34	NDUFS5:92
tRNA Modification (GO:0006400)	-0.19398202	66	5.227e-08	1.759e-05	THADA:6	TRMT10C:27
Potassium Ion Transport (GO:0006813)	0.14504949	115	8.255e-08	2.615e-05	KCNQ3:8	KCNV2:43
Mitochondrial Translation (GO:0032543)	-0.15838148	95	1.007e-07	3.012e-05	PTCD3:108	MRLP3:17
Calcium Ion Transport (GO:0006816)	0.15041269	101	1.853e-07	4.990e-05	CRACR2B:46	CACNB2:52
Mitochondrial ATP Synthesis Coupled Elec	-0.20948974	52	1.778e-07	4.990e-05	NDUFS3:34	NDUFS5:92
Metal Ion Transport (GO:0030001)	0.12506045	145	2.173e-07	5.319e-05	SLC17A4:115	KCNF1:147
Regulation Of Synaptic Transmission, Glu	0.22121363	46	2.139e-07	5.319e-05	CACNG5:152	SHANK3:184
Regulation Of DNA-templated Transcriptio	-0.04094609	1447	4.435e-07	1.038e-04	ZSCAN20:28	BRCA1:66
Aerobic Electron Transport Chain (GO:001	-0.20023570	52	6.022e-07	1.351e-04	NDUFS3:34	NDUFS5:92
Anterograde Trans-Synaptic Signaling (GO	0.11101869	168	7.480e-07	1.611e-04	KCNQ3:8	CACNB2:52
Cardiac Conduction (GO:0061337)	-0.21706995	40	1.243e-06	2.575e-04	CACNB2:52	DSC2:225
Mitochondrial RNA Metabolic Process (GO:	-0.31003253	19	2.906e-06	5.795e-04	POLRMT:30	TFB2M:45
pRNA Processing (GO:0034587)	0.32622871	17	3.226e-06	6.079e-04	PWILL1:2	MOV10L1:6
Translation (GO:0006412)	-0.10289002	173	3.274e-06	6.079e-04	NARS1:76	PTCD3:108
Inorganic Cation Import Across Plasma Me	0.23170854	93	4.610e-06	8.275e-04	SCNN1G:183	CACNA1B:282
Regulation Of Heart Rate By Cardiac Cond	0.12262099	35	5.764e-06	1.001e-03	CACNB2:52	DSC2:225
Cell Junction Assembly (GO:0034329)	0.14489373	80	7.686e-06	1.293e-03	SDK2:26	DBNL:69
tRNA Methylation (GO:0001510)	-0.16607659	57	1.161e-05	1.892e-03	THADA:6	TRMT10C:27
Sensory Perception Of Light Stimulus (GO	0.13358603	90	1.226e-05	1.892e-03	CRYGA:105	WFS1:173
Visual Perception (GO:0007601)	0.13430828	89	1.230e-05	1.892e-03	CRYGA:105	WFS1:173
Cell-Cell Junction Organization (GO:0045	0.15674747	64	1.479e-05	2.212e-03	PKP3:288	GJC1:400
Cellular Respiration (GO:0045333)	-0.15179197	68	1.538e-05	2.238e-03	NDUFS3:34	NDUFS5:92
Regulation Of Transcription By RNA Polym	-0.03432493	1500	1.720e-05	2.438e-03	TET1:5	ZSCAN20:28
Muscle Contraction (GO:0006936)	0.13628952	83	1.822e-05	2.515e-03	MYH13:5	MYH2:19
Neuron Projection Morphogenesis (GO:0048	0.11253550	121	1.993e-05	2.683e-03	DBNL:69	TRIO:114

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_NEURONAL_SYSTEM	0.13583090	345	5.828e-18	3.780e-14	KCNQ3:8
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	-0.31371317	47	1.022e-13	3.314e-10	NDUFS3:34
REACTOME_COMPLEX_I_BIOGENESIS	-0.30113481	48	5.388e-13	1.165e-09	NDUFS3:34
FISCHER_DREAM_TARGETS	-0.07366194	802	2.134e-12	3.460e-09	KIF18B:7
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.21705506	79	2.651e-11	3.439e-08	LRPPRC:1
REACTOME_KERATINIZATION	0.20984294	83	4.015e-11	4.341e-08	DSG3:103
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	0.11967217	255	5.316e-11	4.926e-08	SSTR4:57
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.19105521	98	6.593e-11	5.346e-08	LRPPRC:1
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.30113335	38	1.346e-10	9.702e-08	TTYH3:68
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.08205396	508	3.090e-10	2.005e-07	KCNQ3:8
BENPORATH_ES_WITH_H3K27ME3	0.06242213	898	3.433e-10	2.025e-07	KCNQ3:8
WP_GPCRS_CLASS_A_RHODOPSINLIKE	0.12732830	203	4.363e-10	2.358e-07	SSTR4:57
REACTOME_SENSORY_PERCEPTION	0.10902447	277	4.792e-10	2.391e-07	CACNB2:52
JOHNSTONE_PARVB_TARGETS_3_DN	-0.06948614	689	6.932e-10	3.212e-07	MIS18BP1:17
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	0.20233458	76	1.099e-09	4.754e-07	DSG3:103
YOSHIMURA_MAPK8_TARGETS_UP	0.05562861	1052	1.730e-09	7.014e-07	KCNQ3:8
KEGG_CALCIIUM_SIGNALING_PATHWAY	0.13652673	161	2.406e-09	9.183e-07	P2RX2:86
REACTOME_SIGNALING_BY_GPCR	0.07176626	596	2.777e-09	1.001e-06	SSTR4:57
REACTOME_MUSCLE_CONTRACTION	0.13054939	172	3.728e-09	1.272e-06	NEB:27
MARSON_BOUND_BY_E2F4_UNSTIMULATED	-0.07073743	602	3.928e-09	1.274e-06	KIF18B:7
REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R	-0.13993940	145	6.316e-09	1.951e-06	LRPPRC:1
WP_OXIDATIVE_PHOSPHORYLATION	-0.24698494	46	6.893e-09	2.032e-06	NDUFS3:34
REACTOME_VOLTAGE_GATED_POTASSIUM_CHANNEL	0.25423681	41	1.790e-08	5.049e-06	KCNQ3:8
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	0.14843840	118	2.652e-08	7.169e-06	HRH3:78
MIKKELSEN_MCV8_HCP_WITH_H3K27ME3	0.08165093	390	3.574e-08	9.274e-06	KCNQ3:8
MARTENS_TRETINOIN_RESPONSE_UP	0.06371599	641	4.648e-08	1.160e-05	GJB4:9
DDDD_NASOPHARYNGEAL_CARCIOMA_DN	-0.04916409	1109	4.969e-08	1.194e-05	LRPPRC:1
REACTOME_POTASSIUM_CHANNELS	0.17186718	84	5.306e-08	1.229e-05	KCNQ3:8
REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY	0.10576952	210	1.355e-07	3.032e-05	CACNB2:52
RODRIGUES_THYROID_CARCIOMA_POORLY_DIFFE	-0.06672362	529	1.798e-07	3.888e-05	LRPPRC:1
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.08631997	305	2.375e-07	4.969e-05	HRH3:78
WP_CALCIIUM_REGULATION_IN_CARDIAC_CELLS	0.13819431	115	3.165e-07	6.417e-05	GJB4:9
REACTOME_INTEGRATION_OF_ENERGY_METABOLIS	0.16610491	79	3.391e-07	6.666e-05	CACNB2:52
REACTOME_GPCR_LIGAND_BINDING	0.07500770	385	4.888e-07	9.325e-05	SSTR4:57
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.15149083	91	6.028e-07	1.117e-04	PTCD3:108
REACTOME_G_ALPHA_I_SIGNALING_EVENTS	0.09205190	245	7.455e-07	1.343e-04	SSTR4:57
GEORGES_TARGETS_OF_MIR192_AND_MIR215	-0.05341443	751	7.860e-07	1.378e-04	DS1:5
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	0.36773311	15	8.183e-07	1.397e-04	P2RX2:86
REACTOME_CLASS_A_1_RHODOPSIN LIKE_RECEPT	0.08351310	293	9.486e-07	1.578e-04	SSTR4:57
WP_ELECTRON_TRANSPORT_CHAIN_OXPHOS_SYSTE	-0.16593757	71	1.354e-06	2.196e-04	NDUFS3:34

DisSINET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Mitochondrial CSE	-0.11573909	333	5.447e-13	5.342e-09	LRPPRC:1
Increased CSF lactate	-0.21029171	52	1.593e-07	7.811e-04	LRPPRC:1
Schizophrenia	0.03993086	1503	5.307e-07	1.735e-03	TRANK1:3
Autism Spectrum Disorders	0.06680091	451	1.430e-06	3.506e-03	CACNB2:52
Color vision defect	0.20451491	41	5.943e-06	1.151e-02	OFD1:40
Congenital cataract	0.13798573	89	7.041e-06	1.151e-02	CRYGA:105
Color vision defect, severe	0.21023990	36	1.283e-05	1.260e-02	OFD1:40
Abnormal color vision	0.21023990	36	1.283e-05	1.260e-02	OFD1:40
Loss in color vision	0.21023990	36	1.283e-05	1.260e-02	OFD1:40
NADH:(1) Oxidoreductase deficiency	-0.25215086	25	1.285e-05	1.260e-02	NDUFS3:34
Difficulties with night vision	0.11886660	109	1.876e-05	1.672e-02	KCNV2:43
Acute necrotizing encephalopathy	-0.27694047	19	2.938e-05	2.216e-02	NDUFS3:34
Lactic acidemia	-0.11162049	118	2.923e-05	2.216e-02	LRPPRC:1
Night Blindness	0.11200710	114	3.743e-05	2.621e-02	KCNV2:43
CATARACT, COPPOCK-LIKE	0.39509678	9	4.056e-05	2.652e-02	CRYBB1:478
MITOCHONDRIAL COMPLEX I DEFICIENCY	-0.23067691	26	4.705e-05	2.884e-02	NDUFS3:34
Psychoses, Substance-Induced	0.20365720	15	5.245e-05	3.008e-02	RGS9:188
Mitochondrial Respiratory Chain Deficien	-0.19705510	35	5.522e-05	3.008e-02	PNPT1:137
Psychoses, Drug	0.26233965	19	7.560e-05	3.902e-02	RGS9:188
Huntington Disease	0.05207544	496	8.376e-05	4.107e-02	SH3BP2:30
Abnormal mitochondria in muscle tissue	-0.22804616	25	9.192e-05	4.292e-02	NDUFS3:34
Acidosis, Lactic	-0.09229570	145	1.300e-04	4.553e-02	LRPPRC:1
Night blindness, congenital stationary	0.23064200	23	1.293e-04	4.553e-02	USP11:85
Channelopathies	0.16801221	44	1.165e-04	4.553e-02	SHANK3:184
Progressive macrocephaly	-0.23660039	22	1.227e-04	4.553e-02	NDUFS3:34
Severe myopia	0.11952071	88	1.091e-04	4.553e-02	PDE6B:121
Substance abuse problem	0.09623584	134	1.244e-04	4.553e-02	PWILL1:2
Sudden Cardiac Death	0.16461331	46	1.134e-04	4.553e-02	CACNB2:52
Cataract, Puerulent	0.36751387	9	1.347e-04	4.554e-02	CRYBB1:478
Abnormality of macular pigmentation	0.23478913	21	1.963e-04	6.095e-02	PDE6B:121
Bipolar Disorder	0.04260830	671	2.035e-04	6.095e-02	TRANK1:3
Macular pigmentary changes	0.23478913	21	1.963e-04	6.095e-02	PDE6B:121
Nuclear non-senile cataract	0.19926817	29	2.051e-04	6.095e-02	WFS1:173
Autistic Disorder	0.04477948	576	2.811e-04	7.984e-02	FRMPD4:45
Growth retardation	-0.08688568	146	3.011e-04	7.984e-02	MAN2B1:13
Pfaundler-Hurler Syndrome	-0.21319195	24	3.013e-04	7.984e-02	MAN2B1:13
Poor growth	-0.08688568	146	3.011e-04	7.984e-02	MAN2B1:13
Nuclear cataract	0.18720716	31	3.111e-04	8.027e-02	WFS1:173
Growth delay	-0.08622058	147	3.196e-04	8.037e-02	MAN2B1:13
Electroretinogram abnormal	0.09283442	123	3.880e-04	9.513e-02	OFD1:40

MGI_Mammalian_Phenotype_Level_4 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0003635 abnormal synaptic transmissio	0.08281771	381	7.372e-08	3.509e-05	IQSEC1:33
MP0005551 abnormal eye electrophysiology	0.10983361	134	1.396e-05	3.322e-03	CACNB2:52
MP0002085 abnormal embryonic tissue	-0.05265556	583	3.484e-05	5.529e-03	FUZ:22
MP0002086 abnormal extraembryonic tissu	-0.05826971	429	6.411e-05	6.103e-03	CHRD:40
MP0002429 abnormal blood cell	-0.03942756	1124	5.540e-05	6.103e-03	MAN2B1:13
MP0002572 abnormal emotion/affect behav	0.07039395	273	8.957e-05	7.106e-03	PLA2G6:77
MP0002066 abnormal motor capabilities/c	0.03963142	982	1.150e-04	7.821e-03	LPIN1:4
MP0002971 abnormal brown adipose	0.13264024	70	1.351e-04	8.040e-03	LPIN1:4
MP0005253 abnormal eye physiology	0.11193102	87	3.375e-04	1.344e-02	CACNB2:52
MP0000716 abnormal immune system	-0.03788023	917	3.298e-04	1.344e-02	MAN2B1:13
MP0002733 abnormal sensory capabilities	0.06885800	237	3.387e-04	1.344e-02	LPIN1:4
MP0002064 seizures	0.07047796	230	2.990e-04	1.344e-02	KCNQ3:8
MP0002998 abnormal bone marrow	-0.04063861	743	3.993e-04	1.462e-02	DST:15
MP0000689 abnormal spleen morphology	-0.04678284	506	5.486e-04	1.742e-02	MAN2B1:13
MP0001970 abnormal pain threshold	0.08458003	142	5.781e-04	1.742e-02	SSTR4:57
MP0001697 abnormal embryo size	-0.05409887	363	5.854e-04	1.742e-02	CHRD:40
MP0002063 abnormal learning/memory/cond	0.05640510	325	6.596e-04	1.847e-02	NETO1:50
MP0003935 abnormal craniofacial develop	-0.06770459	216	7.479e-04	1.978e-02	FUZ:22
MP0000548 abnormal adipose tissue	0.05565087	292	1.392e-03	3.487e-02	LPIN1:4
MP0000592 abnormal liver morphology	-0.04558479	440	1.561e-03	3.710e-02	MAN2B1:13
MP0005380 embryogenesis phenotype	-0.04775063	367	2.288e-03	3.710e-02	NACA:32
MP0002735 abnormal chemical nociception	0.16739411	28	2.217e-03	3.710e-02	ADCY1:94